

Kollepara Naga Sai Manoj Abhiram

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SUMMARY

Software Engineer with production experience in backend development and Generative AI. Proficient in Go and Python, with hands-on expertise building microservices, REST APIs, and Kafka-driven event-driven systems across CRM and EdTech platforms. Experienced in OCR pipelines, RAG workflows, Agentic AI, and domain-specific LLM fine-tuning. Passionate about building scalable, production-grade systems at the intersection of software engineering and AI.

EDUCATION

Shiv Nadar University Chennai

B.Tech Computer Science and Engineering (Internet of Things)

Chennai, Tamil Nadu

Sept 2021 – June 2025

EXPERIENCE

Deepta AI

Software Engineer 1

Hyderabad, Telangana

Aug 2025 – Present

- Designed OCR-based document processing pipelines for CRM and EdTech platforms, turning unstructured inputs into structured data and cutting manual validation overhead by ~60%.
- Engineered Kafka-driven event pipelines in Go to stream real-time data across microservices, enabling high-throughput email and WhatsApp notifications for 10K+ users.
- Configured and scaled Omnichannel engagement platforms to orchestrate campaigns across email and WhatsApp, improving reach across CRM and EdTech products.
- Strengthened authentication reliability by building user verification flows and Redis-backed session management, cutting authentication latency by ~40%.
- Shipped 15+ REST APIs and CRUD modules in Go and PostgreSQL, powering listing pages, user interactions, and data management across CRM and EdTech platforms.
- Built Generative AI features including RAG pipelines and Agentic AI workflows, experimenting with LLM integrations and domain-specific fine-tuning for CRM use cases.

PROJECTS

Pest Detection Application | Deep Learning, Flask, React, AWS EC2, Selenium, Python

- Scraped and labeled a custom pest image dataset of 5K+ images across different classes from BugGuide using Selenium, training an Inception-based CNN with augmentation to address severe class imbalance.
- Deployed a Flask + React app on AWS EC2 achieving ~75% classification accuracy with real-time pest identification from uploaded images.

Automated Personal Finance Manager | MERN Stack, Python, Machine Learning, NLP

- Built a full-stack finance platform with ML-based transaction categorization using Random Forest achieving ~85% accuracy, and an NLP chatbot for financial Q&A.
- Delivered 6+ features including expense forecasting, interactive dashboards, budget alerts, multi-currency support, and a risk-profile-based stock recommendation module.

TECHNICAL SKILLS

- Programming Languages:** Go, Python, SQL
- Frameworks & Tools:** Kafka, Microservices, REST APIs, Flask, FastAPI, Streamlit, React, Selenium, Git, Postman
- Databases & Caching:** PostgreSQL, Redis, MongoDB
- AI / GenAI:** OCR, OpenCV, NLP, LLM Fine-Tuning, Prompt Engineering, RAG, Agentic AI, Vector Databases

CERTIFICATIONS

- IBM Generative AI Engineering Professional Certificate – Coursera
- Building Microservices API in Go – Coursera
- Joy of Computing using Python – NPTEL
- Data Mining – NPTEL